University of Sunderland

Application 035907

Section A: Applicant details

Date application started:
Thu 5 March 2026 at 10:13

First name:
Singo

Last name:
Loua

Email:
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Programme name:
MSc Applied Data Science

Module name:
PROM02 MSc Dissertation
Last updated:
06/03/2026

Department:
School of Computer Science

Applying as:
Student

Research project title:
Deep Learning and Visual Language Model (VLM) based Real-Time Fighting Detection on Beach Safety Management

Similar applications:
- not entered -

Supervisor

1. Supervisor

Name	Email
Dr Ming Jiang	Ming.Jiang@sunderland.ac.uk

Risk Assessment

Suitability

Adheres to BPS Code of Human Research Ethics (2021)?
No

Takes place outside UK?
No

Involves NHS?
No

Healthcare research?
No

ESRC funded?
No

Involves adults who lack the capacity to consent?
No

Led by another UK institution?
No

Involves human tissue?
No

Clinical trial?
No

Social care research?
No

Risk Assessment

Does the study involve participants who are potentially or in any way vulnerable or who may have any difficulty giving meaningful consent to their participation or the use of their information?
No

Are participants to be involved in the study without their knowledge and consent (e.g. through internet-mediated research, or via covert observation of people in public places)?
No

Will the study require the co-operation of a gatekeeper for initial access to the groups or individuals to be recruited?
No

Does the research methodology involve the use of deception or activities which are conducted without participants' full and informed consent at the time the study is carried out?
No

Are there any significant concerns regarding the design of the research project?
No

Does the research involve any of the following groups?

a. children under 18 years of age?
b. vulnerable adults (eg people with learning or communication difficulties)
c. individuals who have a dependent or subordinate relationships to researchers
d. people in custody (eg young offenders or people in prisons)
e. individuals unable to give consent
f. individuals involved in illegal activities
g. therapeutic interventions

No

If the proposed research relates to the provision of social or human services is it feasible and/or appropriate that service users or service user representatives should be in some way involved in or consulted?
No

Are there payments to researchers/participants that may have an impact on the objectivity of the research?
No

Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?
No

Could the study induce unacceptable psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life? Will the study involve prolonged or repetitive testing?
No

Will the study involve discussion of sensitive topics? For example (but not limited to): sexual activity, illegal behaviour, experience of violence or abuse, drug use, etc.)
Yes
Are drugs, placebos or other substances to be administered to the study participants or will the study involve invasive, intrusive or potentially harmful procedures of any kind?
No

Will research involve the sharing of data or confidential information beyond the initial consent given?
No

Is there ambiguity about whether the information/data you are collecting is considered to be public?
No

Will the research involve administrative or secure data that requires permission from the appropriate authorities before use?
No

Will the research involve the use of visual/vocal methods that potentially pose an issue regarding confidentiality and anonymity?
Yes

The Data Protection Act 2018 will apply to any data-processing activities entailed by this research. Is there any cause for uncertainty as to whether the research will fully comply with the requirements of the Act?
No

Are there any particular groups who are likely to be harmed by dissemination of the results of this project?
No

Do you have any doubts or concerns regarding your (or your colleagues) physical or psychological wellbeing during the research period?
No

Will the research involve accessing security-sensitive material, such as material related to terrorism or to violent extremism of any kind, including, but not limited to, Islamist extremism and far-right extremism?
No

Summary

This research investigates the effectiveness of integrating YOLOv8 with Visual Language Model (VLM) agents (specifically Gemini/LLaVA) for the real-time detection of physical altercations in beach safety management. The central research question asks: Can a VLM-enhanced computer vision system accurately detect fighting and facilitate natural language tracking while preserving individual privacy through thermal silhouette rendering?
The study does not involve live human participants. Instead, it utilizes secondary data from established academic datasets, including UCF-Crime and RWF-2000, consisting of approximately 600–1,000 video clips. Additionally, the project will utilize synthetic data generation via Google Gemini to augment the dataset with beach-specific edge cases.
The methodology follows a deep learning pipeline: raw video data is pre-processed into anonymized thermal silhouettes to comply with UK GDPR data minimization. This data is used to train a YOLOv8 model for action detection, which is then integrated with a VLM agent to enable security personnel to query surveillance feeds using natural language. Success is measured by achieving a minimum 0.75 mAP and a processing speed of 25 FPS.

Section B: Basic information

Proposed project duration

Start date (of data collection):
Thu 5 March 2026

Anticipated end date (of project)
Mon 11 May 2026

Indicators of risk

Involves potentially vulnerable participants?
No
Involves potentially highly sensitive topics?
Yes

Section C: Summary of research

1. Aims & Objectives

Aim:
Developing and evaluating a system for detecting beach fights in real time is the project’s main goal using both YOLOv8 and an AI agent

using Visual Language Models (VLM). This will include a pipeline for rendering thermal silhouettes of people on the beach to protect their privacy and a ChatBot in Natural Language that security personnel will use to ask questions about—and view in real time—people on the beach from the camera.

Objectives:
• Complete a synthesis of ten scholarly journal articles from 2020 through 2026 regarding (1) fight detection (2) vision-language models, e.g., LLaVA, CLIP (3) artificial intelligence agents used in surveillance, (4) thermal-based rendering of private information, and (5) safety systems for beach use. The synthesis will provide an annotated bibliography of the research gaps related to these areas.
• Start collecting 600 to 1,000 video clips of fighting incidents at beaches/parks from the UCF-Crime and RWF-2000 datasets. Once these video clips are collected, use the thermal silhouette rendering pipeline to anonymize all individuals from each of the video clips. Create an annotated data set with 70% training, 15% validation, and 15% testing splits.
• Utilize a YOLOv8 model to train for physical assault detection that is adopted to work in conjunction with a Vision-Language Model (VLM), such as LLaVA or CLIP, for allowing scene-level reasoning. The final model should achieve a minimum mAP at a threshold of 0.5 = 0.75 and demonstrate the ability to perform inference at 25 frames per second (FPS) or faster, when evaluated on 720p beach surveillance video.
• Evaluate the entire model system using the test data set on the beach, employing metrics including Precision, Recall, F1-score, mAP at 0.5, FPS, false positive rate, and accuracy of VLM output. The evaluation will include a comparison of the results to at least one published baseline in the academic literature
• Use the AIF360 or Fairlearn framework to perform an audit of fairness and bias on the demographic groups and beach lighting conditions, then validate that the thermal silhouette rendering meets the requirements of the UK GDPR Article 5 data minimization principles by effectively anonymizing identifiable features.
• Develop and demonstrate a working prototype of beach safety that integrates the YOLOv8, VLM AI agent, thermal silhouette renderer, and natural language chatbot in a Gradio/Streamlit dashboard to enable live operator tracking queries with real-time responses powered by the VLM.

2. Methodology

For this project, I will be using the Cross-Industry Standard Process for Data Mining, method because it is a structured approach with the advantage of allowing me to change it if it does not work, instead of ending up with a poor outcome.

Firstly, I identified the problem, which is the fact that beaches are crowded, and a fight could erupt at any time. In addition, there is no AI system that has been developed to solve this problem with consideration of privacy. I have read six papers on fight detection, visual language models, AI agents in surveillance, thermal privacy, and beach safety, and I will read ten in total. On the 20th of February, 2026, I will meet with my supervisor and finalize the problem, and I will also ensure that I have the following requirements in the project: real-time detection of a fight on a beach, an AI agent, thermal silhouette privacy, and a natural language chatbot.

Next, I will analyze the dataset, which will comprise the UCF-Crime and RWF-2000 datasets, and I will check the quality of the videos, the lighting, and the balance between the fighting and non-fighting videos. I will not collect any videos or data.

Before the AI model is developed, the videos will pass through a self-developed pipeline, which will anonymize the faces, the skin, and the clothing, leaving a body outline that the AI will detect to identify the fights in the videos.

For the model, I will develop a combination of the YOLOv8 and the visual language model, which will reduce false alarms, and I will also develop a natural language chatbot and a DeepSort tracker.

For the evaluation, I will use the Precision, the recall, the F1 score, the mAP@0.5, the FPS, and the false positive rate, and I will also check the fairness audit and the UK GDPR Article 5 audit, and I will also develop a Gradio or a Streamlit prototype dashboard.

There are no human subjects in this study.

3. Personal Safety

Raises personal safety issues?

No

Section D: About the participants

1. Potential Participants

No. This project involves no human participants.

2. Recruiting Potential Participants

No participants will be recruited.

3. Consent

Will informed consent be obtained from the participants? (i.e. the proposed process) No

This project involves no human participants

4. Payment

Will financial/in kind payments be offered to participants? No

5. Potential Harm to Participants

What is the potential for physical and/or psychological harm/distress to the participants?

This project involves no human participants

How will this be managed to ensure appropriate protection and well-being of the participants?

This project involves no human participants

Section E: About the data

1. Data Confidentiality Measures

This project involves no human participants and no personal data is collected. All data used comes from pre-existing, publicly available academic video datasets which will be fully anonymised through a thermal silhouette rendering pipeline before use.

2. Data Storage

All data used in this project comes from pre-existing, publicly available academic video datasets. All project files, code, and results will be stored securely on my personal password-protected device and backed up on the University's secure storage system. Access will be restricted to me and my supervisor only. Upon completion of the project, all raw dataset files will be deleted in line with UK GDPR and the Data Protection Act 2018. This project is not externally funded.

Section F: Supporting documentation

Information & Consent

Participant information sheets relevant to project?
No

Consent forms relevant to project?
No

Additional Documentation

Document 1040793 (Version 1) All versions
Planning Review

Document 1040794 (Version 1) All versions
Project Proposal Form

External Documentation

- not entered -

Section G: Declaration

Signed by:
Singo Loua
Date signed:

Fri 6 March 2026 at 01:15

Official notes

- not entered -